WINDOWS SERVER 2022

ACTIVE DIRECTORY & NETWORK ADMINISTRATION PROJECT

Enterprise Identity, Security, and Policy Management

Prepared by:

Kartikay Dhir

Technologies Used:

Windows Server 2022, Active Directory, DNS, DHCP,

Group Policy, NTFS Permissions, Security Auditing

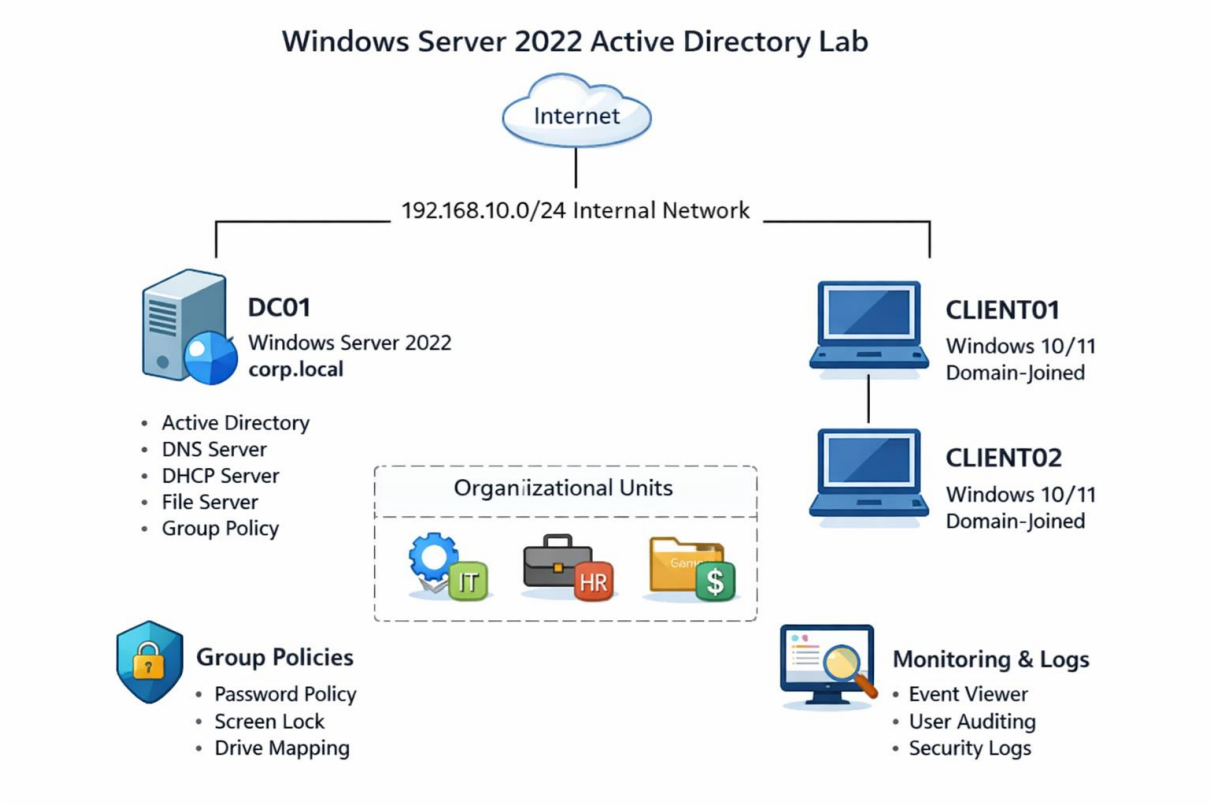
This project demonstrates the deployment, configuration, testing, and troubleshooting of a Windows Server 2022 enterprise environment, including Active Directory Domain Services, Group Policy Objects, secure file access, and security auditing aligned with industry best practices.

Date: 29th January 2026

WINDOWS SERVER 2022 ACTIVE DIRECTORY PROJECT DOCUMENTATION

1. PROJECT OVERVIEW

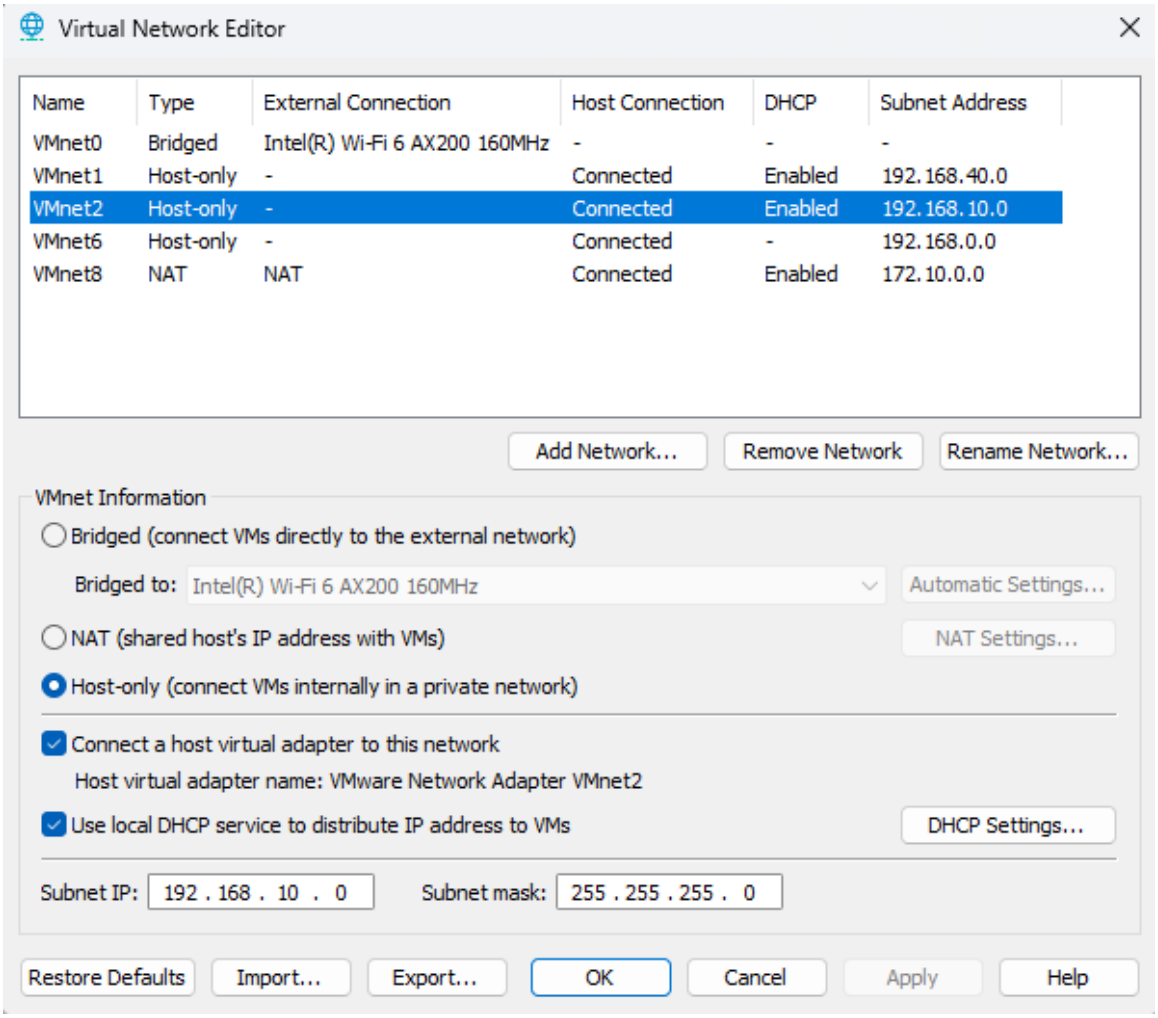
This project demonstrates the deployment and administration of a Windows Server 2022 Active Directory environment, including DNS, DHCP, Group Policy, file services, and security auditing aligned with enterprise and Canadian IT best practices.



2. NETWORK ARCHITECTURE

Domain Name: corp.local
Network: 192.168.10.0/24
Domain Controller: DC01 (192.168.10.10)
Clients: CLIENT01, CLIENT02 (DHCP assigned)

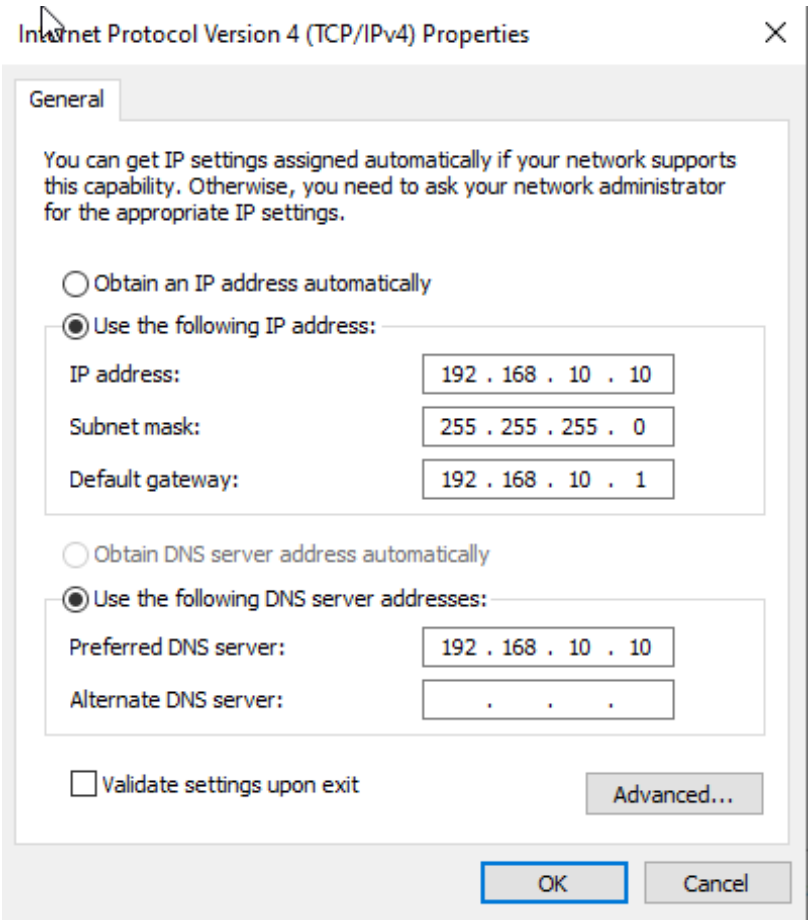
- VMware virtual host network created for this project (**VMnet2**)



3. SERVER INSTALLATION & CONFIGURATION

Windows Server 2022 was installed on a virtual machine and configured with a static IP address. System updates were applied and the server was renamed to DC01.

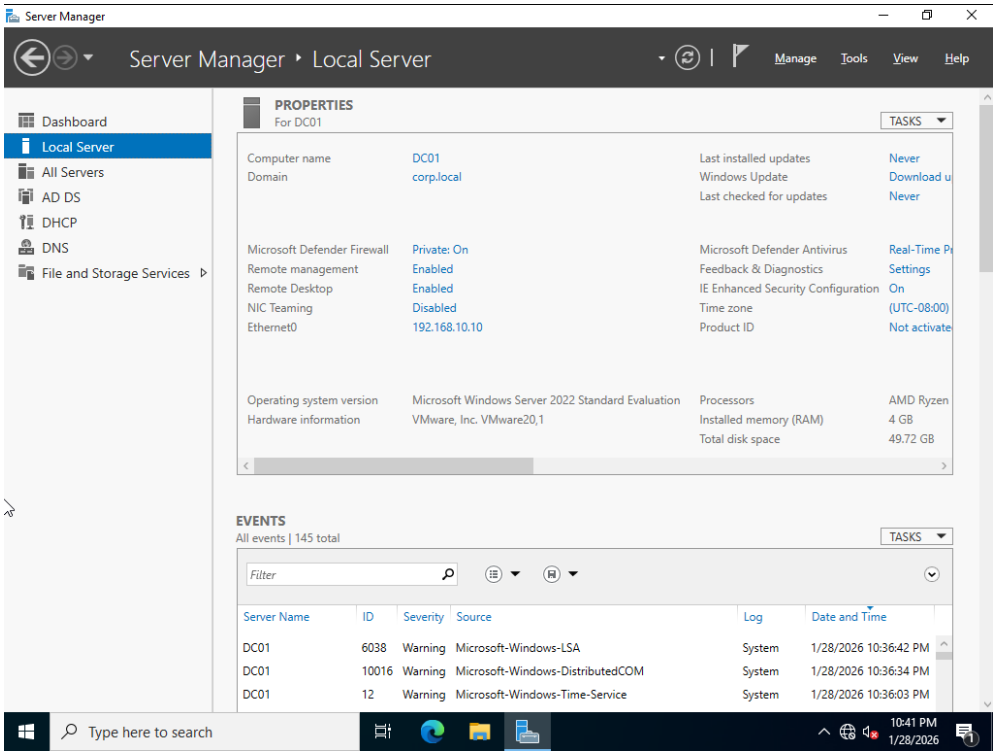
- **Static IPv4 Address** for the Server



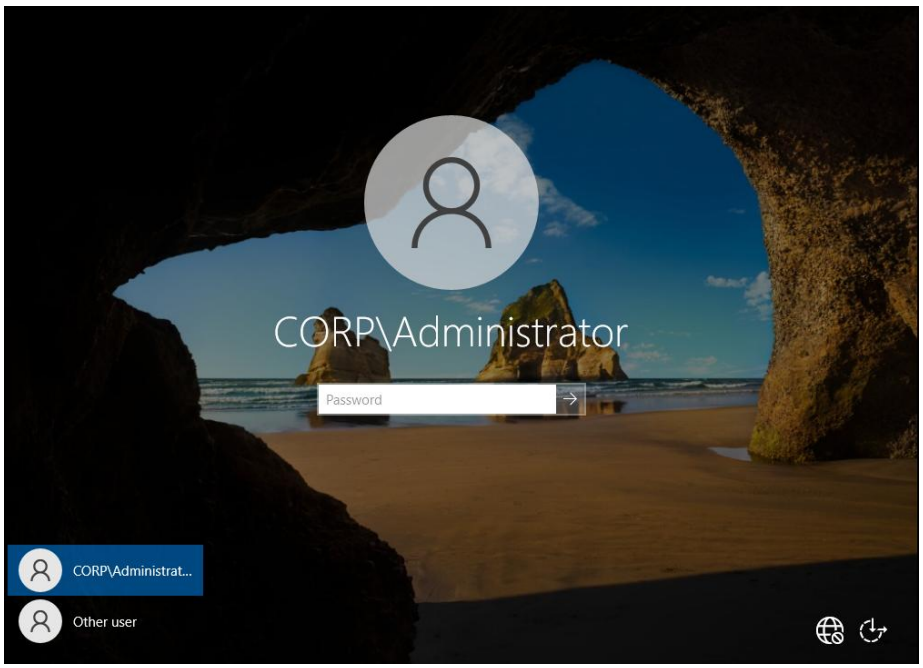
4. ACTIVE DIRECTORY & DNS SETUP

Active Directory Domain Services and DNS were installed. The server was promoted to a Domain Controller using a new forest configuration.

- **AD DS & DNS** Roles installed on the server



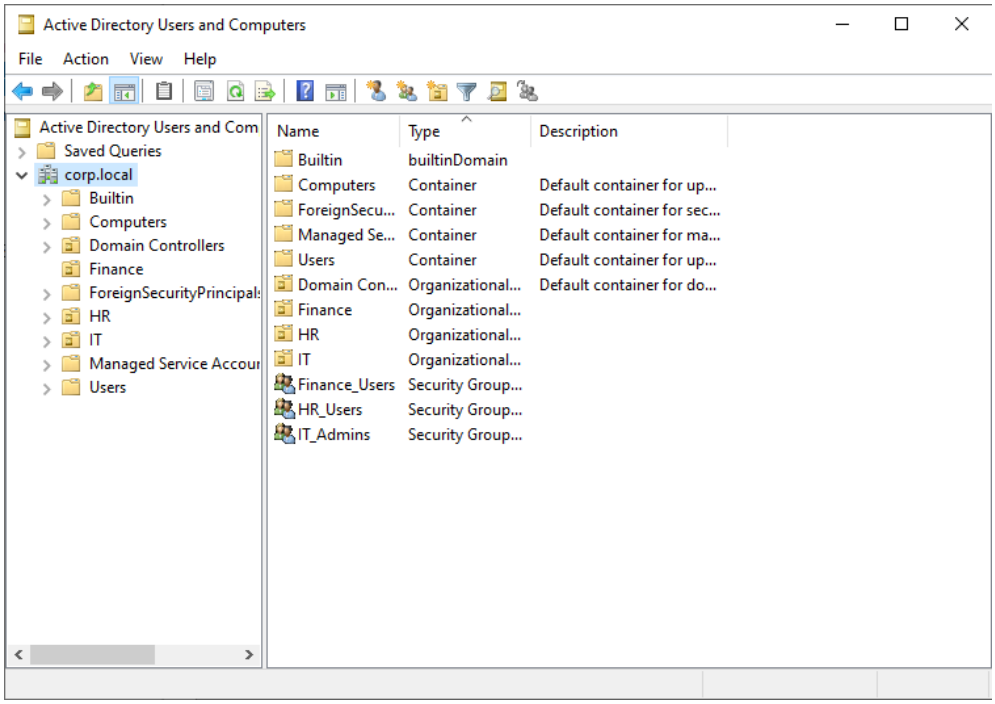
- Server Promoted to **Domain Controller**



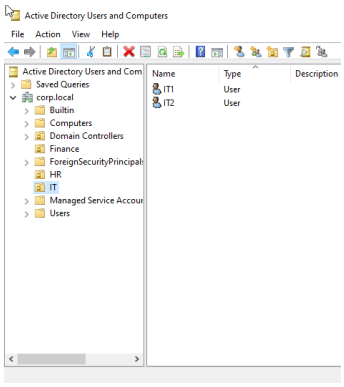
5. ORGANIZATIONAL UNITS & USER MANAGEMENT

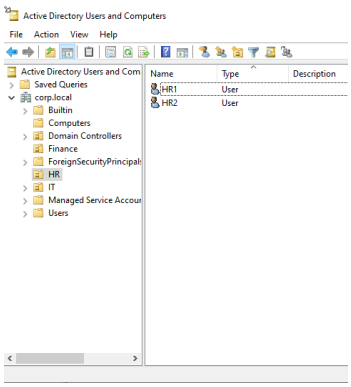
Organizational Units were created to represent departments. Users and security groups were created and assigned using role-based access control.

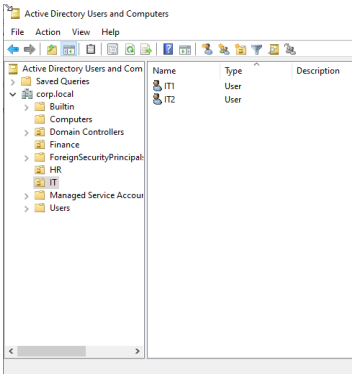
- Organizational Unit Created in the domain **(HR, Finance, IT)**



- **Test Users** created in all Organizational Units



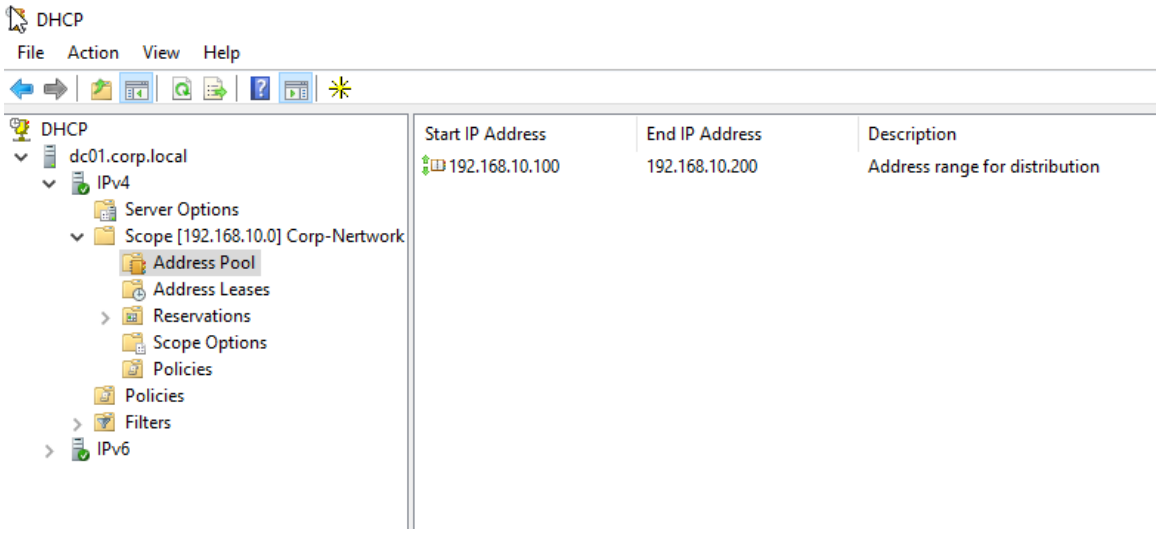




6. DHCP CONFIGURATION

The DHCP Server role was installed and authorized. A DHCP scope was configured to automatically assign IP addresses to client systems.

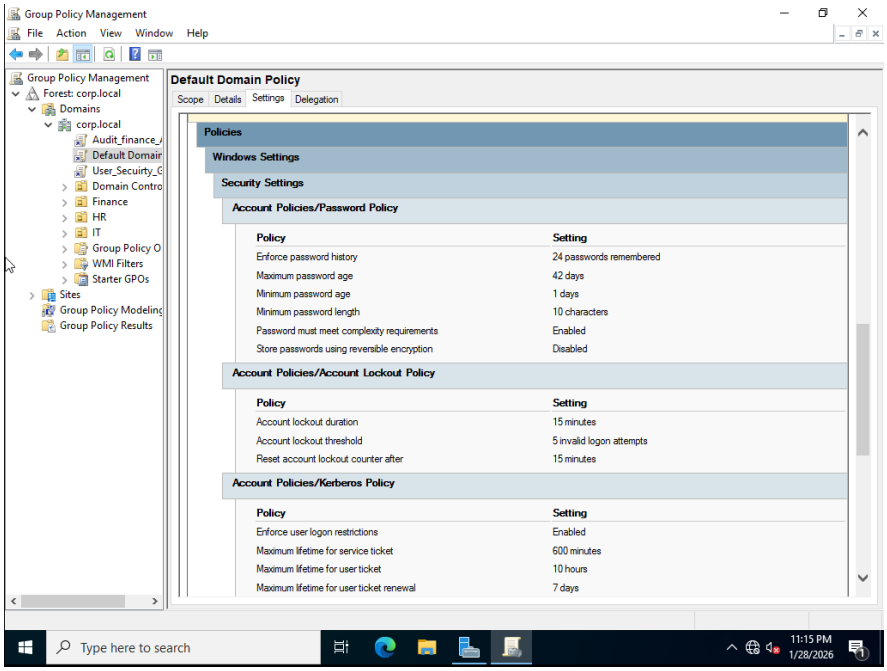
- **DHCP Installed** and New Scope known as **Corp-Network** Created



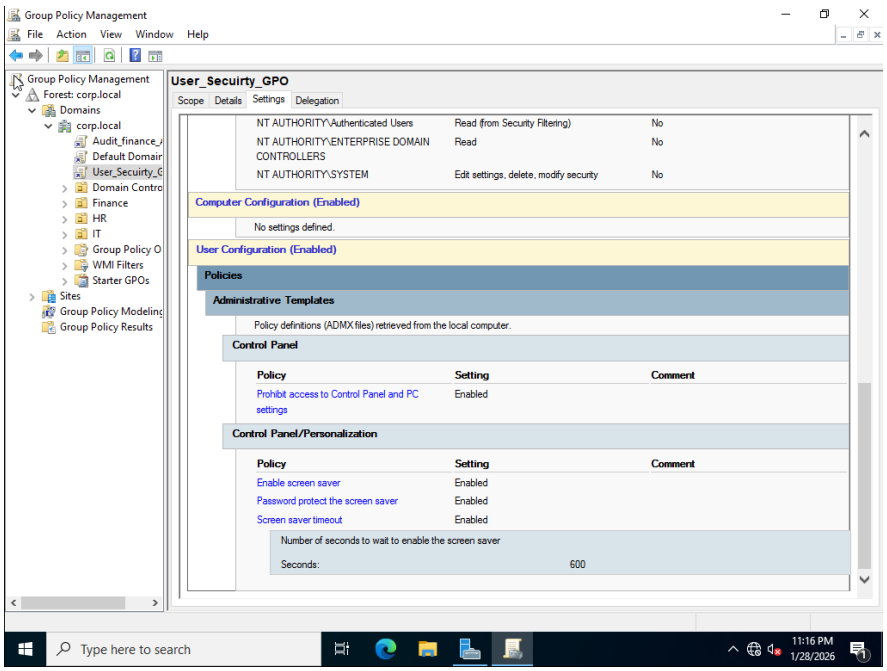
7. GROUP POLICY IMPLEMENTATION

Group Policy Objects were created and linked to enforce password policies, screen lock timers, control panel restrictions, and drive mappings.

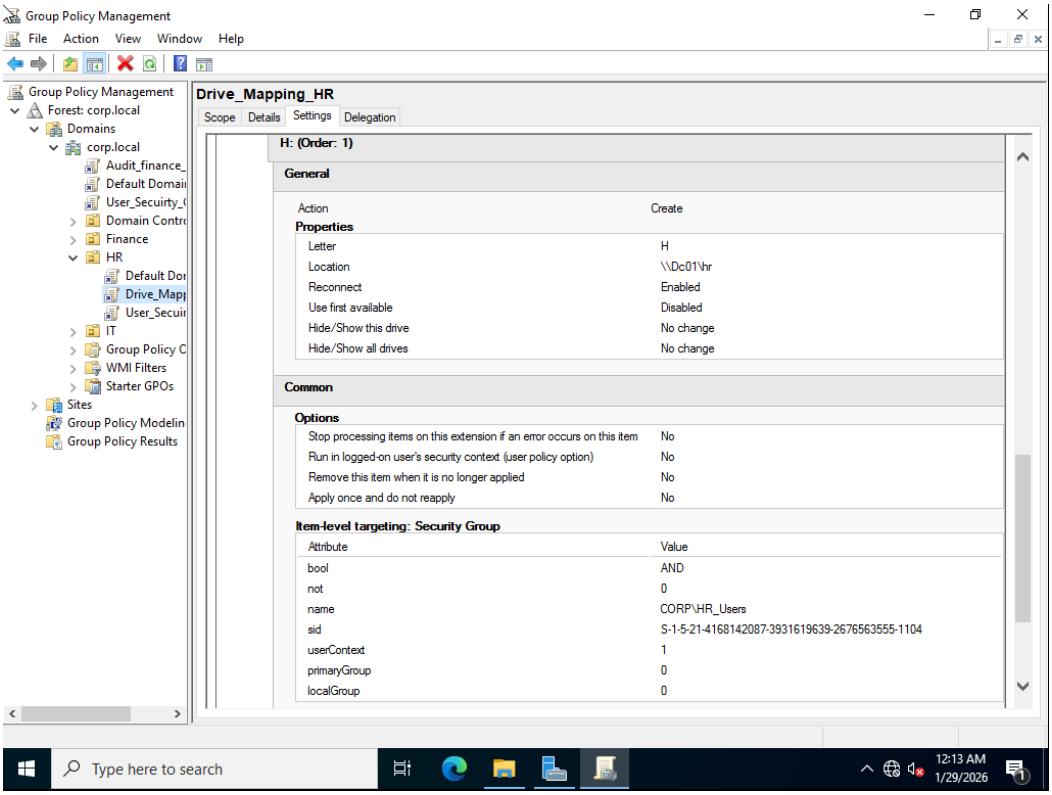
- Made Changes to **Default Domain Policy** to enforce password policies



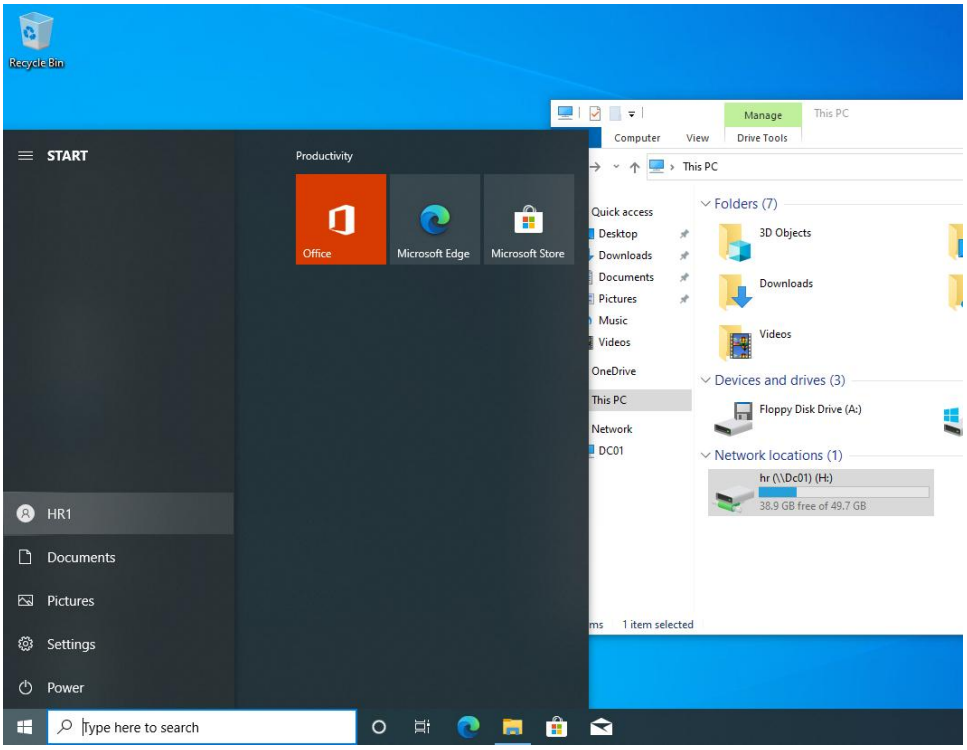
- Created a new group policy called **User_Security_GPO** for unauthorized users to access control panel and set up screen lock timers



- Created a new group policy called **Drive_Mapping_HR** to automatically provide finance user to get access to shared files



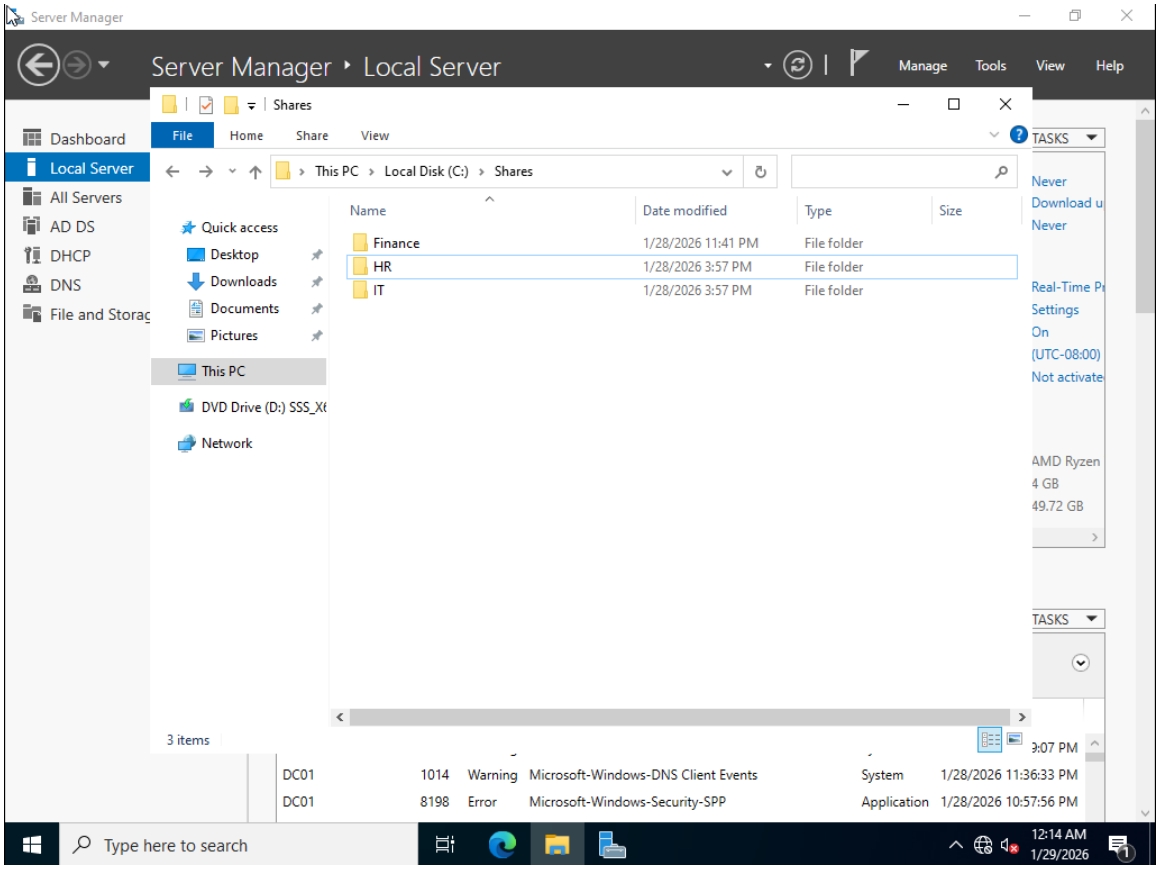
- **Mapped Drive** showing on HR1



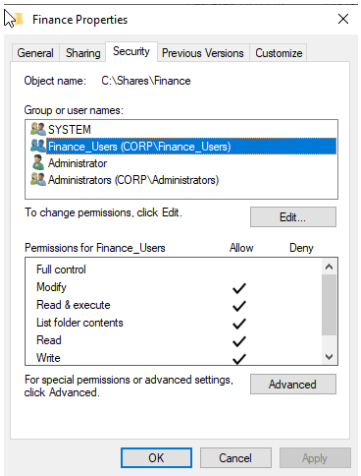
8. FILE SERVER & PERMISSIONS

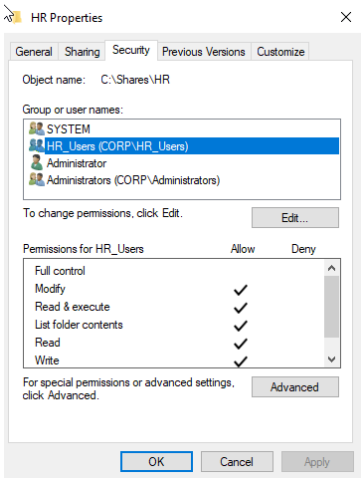
Departmental file shares were created with NTFS permissions applied based on security group membership.

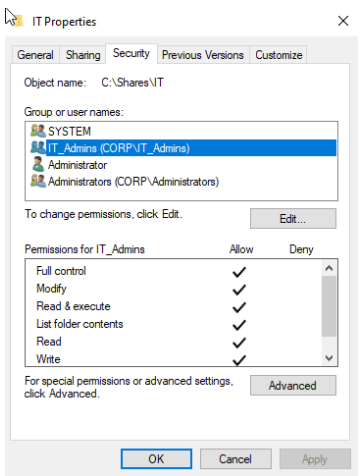
- **File structure** of shared files in DC01



- **NTFS Share Permissions** for all the shared folders



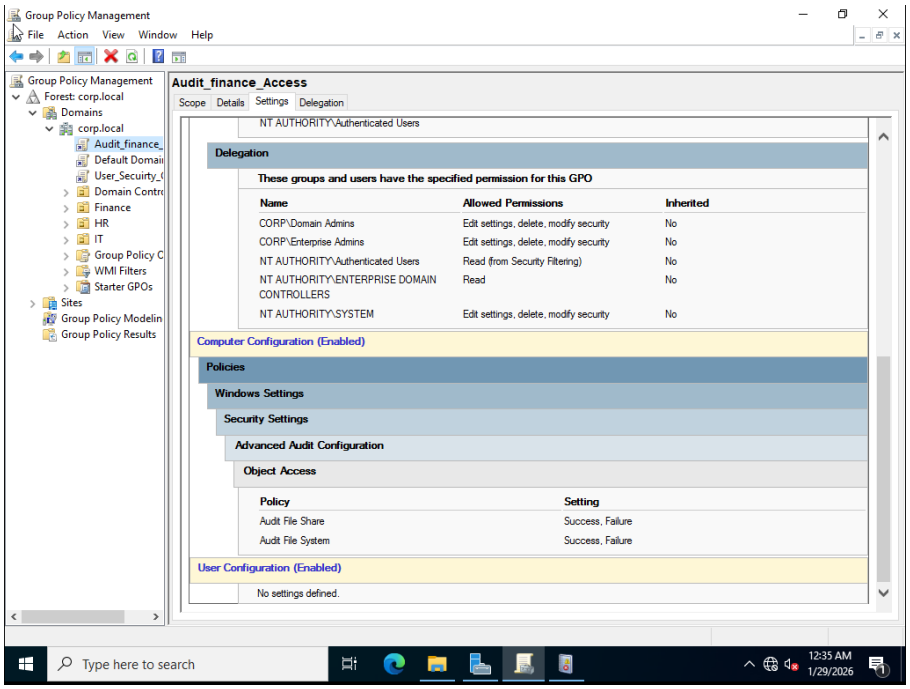




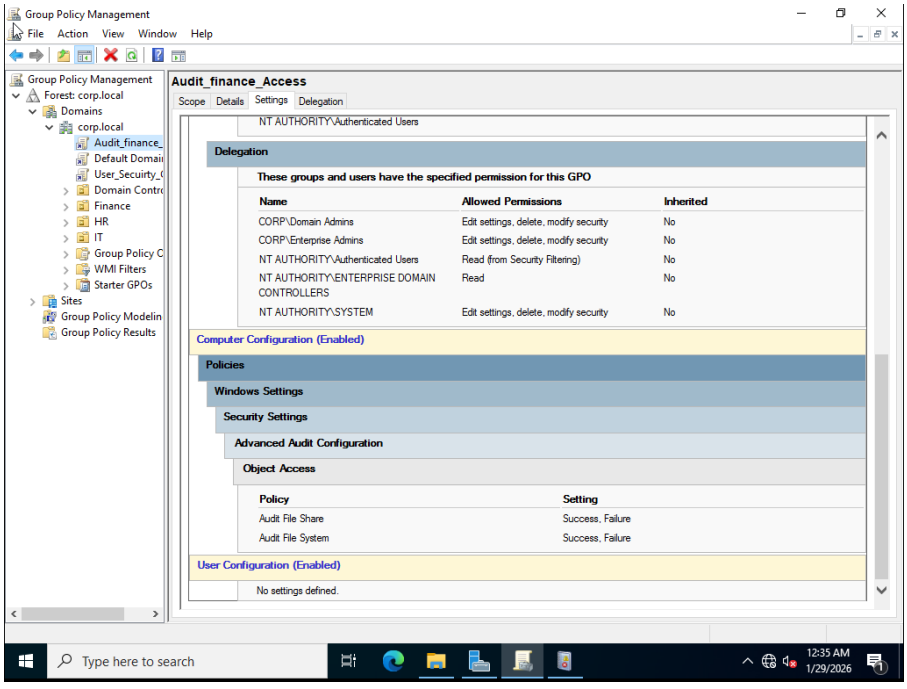
9. SECURITY AUDITING (FINANCE)

Auditing was enabled via Group Policy and configured on the Finance folder to track successful and failed access attempts.

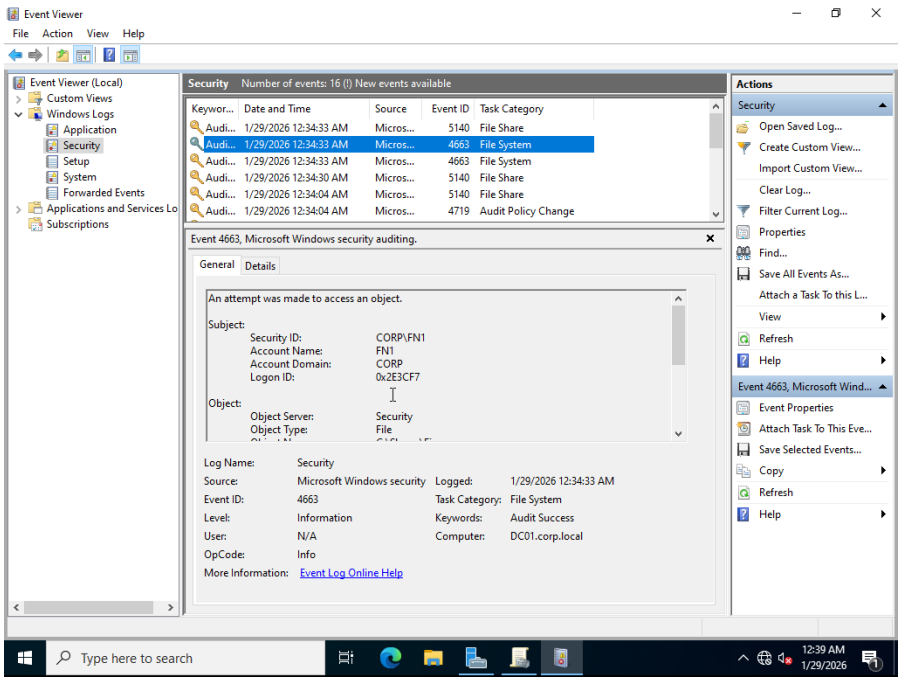
- Create a new group policy called **Audit_finance_Access** to track successful and failed attempts on the shared folder



- **Auditing Settings** on Finance Folder on DC01



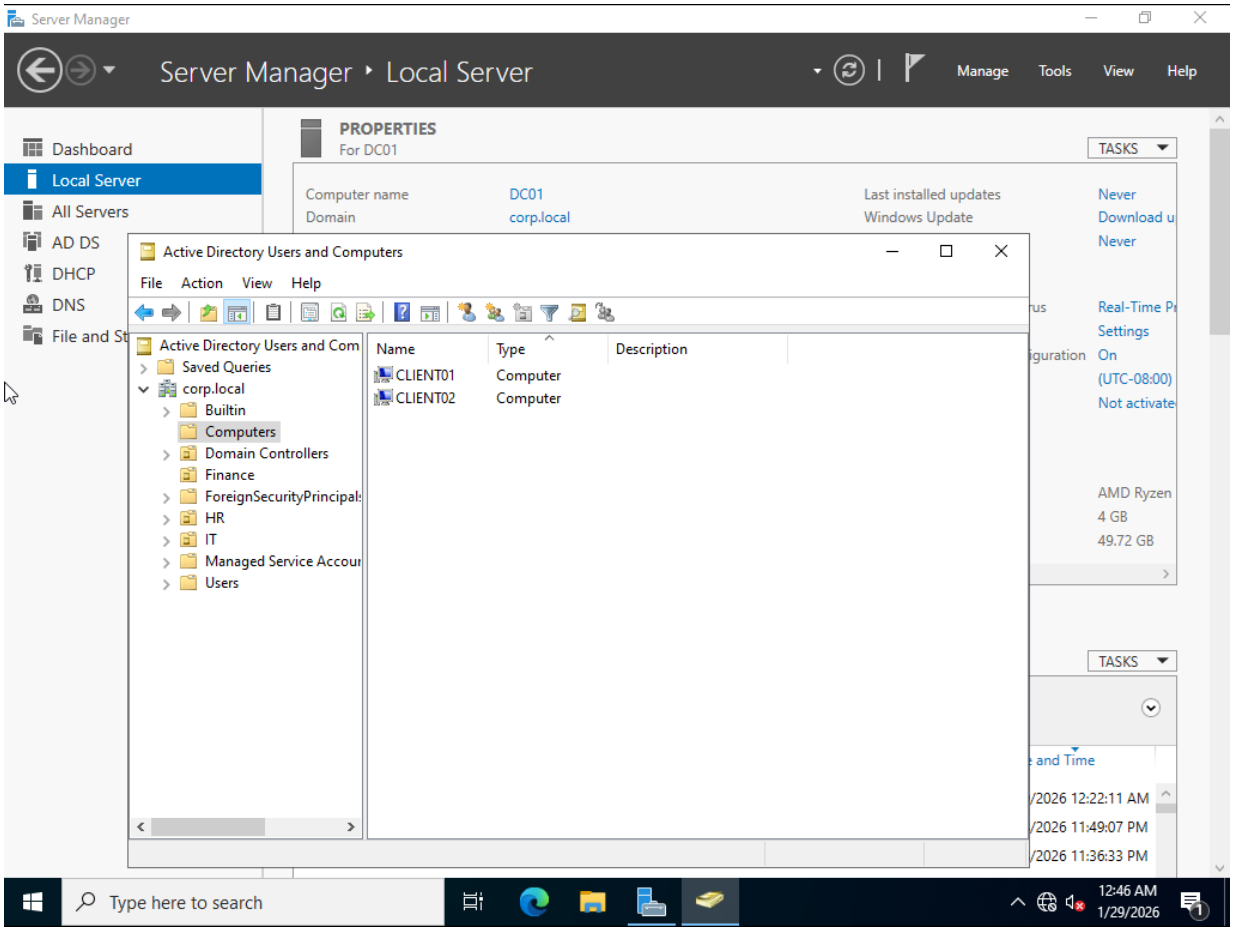
- **Event Log** for when I created a test folder in the finance from FN1



10. CLIENT CONFIGURATION & TESTING

Client machines were joined to the domain and tested for Group Policy application, drive mapping, and file access permissions.

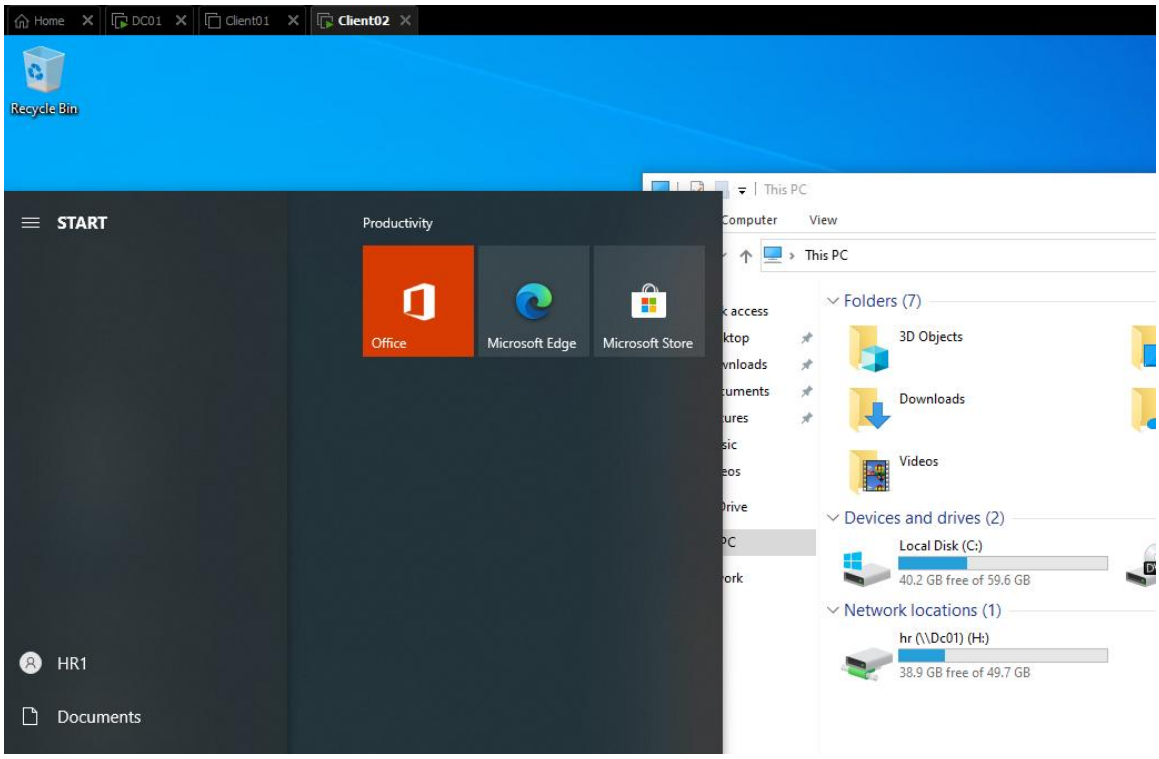
- Both the **Client devices** are joined to the domain



- Used **gpresult /R** to find out what group policies are applied on HR1



- **Drive Mapping** showed on Client02 logged as HR1

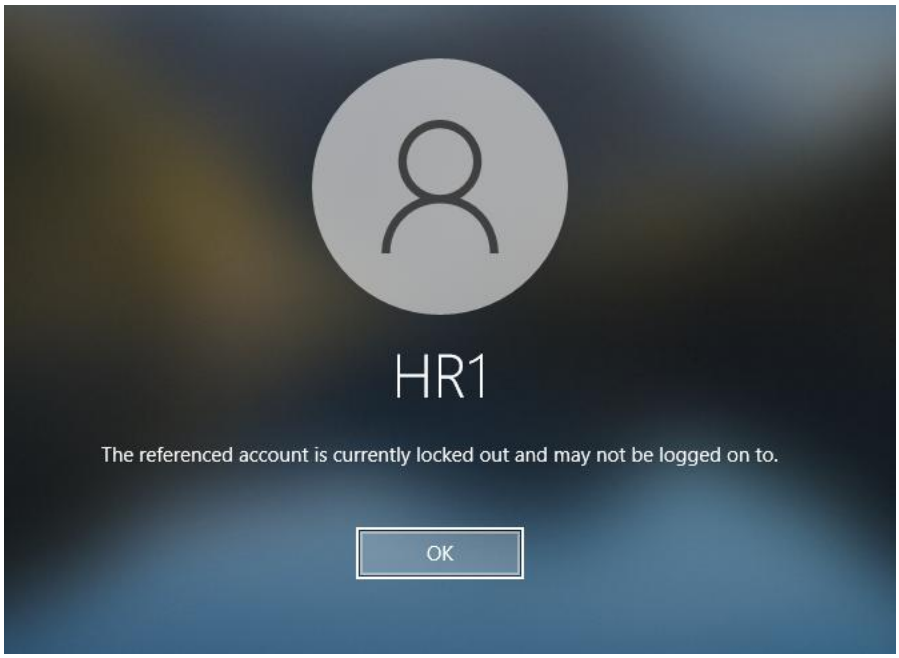


11. TESTING & TROUBLESHOOTING

Simulated incidents were performed including account lockouts and unauthorized access attempts.

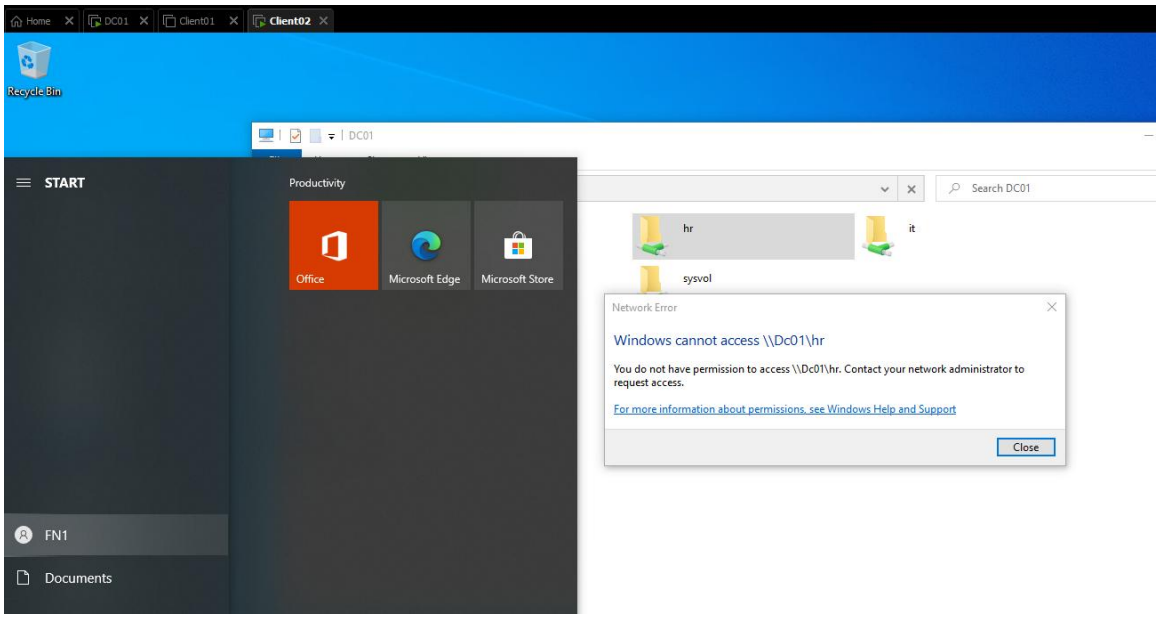
TEST 1: LOCKING THE ACCOUNT

- **Trigger account lock** on Client02 (Unsuccessful attempts: 5)



TEST 2: UNAUTHORIZED ACCESS DENIED

- Logged in as one of the Finance User to try accessing the finance folder



13. ISSUE FACED AND RESOLUTION

This section documents issues encountered during the implementation of the Windows Server 2022 Active Directory environment and the steps taken to resolve them. Documenting these challenges demonstrates practical troubleshooting skills and real-world system administration experience.

ISSUE 1: GROUP POLICY FAILED TO APPLY ON CLIENT MACHINE

Problem: When running **gpupdate /force** on the client computer, Group Policy processing failed with errors indicating that the computer and user names could not be resolved.

Root Cause: The client machine was not configured to use the Domain Controller as its primary DNS server. Group Policy relies on Active Directory and DNS for name resolution, and external DNS servers prevent proper policy processing

Resolution:

- Verified DNS configuration using **ipconfig /all**
- Updated the client's network adapter to use the Domain Controller's IP address as the primary DNS server
- Flushed and re-registered DNS using **ipconfig /flushdns** and **ipconfig /registerdns**
- Successfully reapplied policies using **gpupdate /force**

ISSUE 2: DRIVE MAPPING GPO NOT APPEARING FOR FINANCE USERS

Problem: Finance users did not see the mapped network drive after logging in.

Root Cause: Drive mapping was configured using item-level targeting that required both correct OU placement and security group membership. One or both conditions were not initially met.

Resolution:

- Verified the user account was located in the **Finance OU**
- Confirmed the user was a member of the **Finance_Users security group**
- Forced a Group Policy update and logged out/in
- Verified GPO application using **gpresult /r**

ISSUE 3: FILE ACCESS AUDITING NOT GENERATING EVENT LOGS

Problem: No security events were generated when Finance users accessed files on the server.

Root Cause: Only File Share auditing was enabled. NTFS auditing requires the **Audit File System** subcategory to be enabled using Advanced Audit Policy Configuration. Additionally, NTFS auditing was not configured on the Finance folder itself.

Resolution:

- Enabled **Audit File System** (Success and Failure) via GPO
- Linked the auditing GPO to the **Domain Controllers OU**
- **Enabled NTFS auditing** on the Finance folder for Finance_Users
- Generated file access activity to trigger events
- Verified audit events in **Event Viewer (Event ID 4663)**

12. CONCLUSION

This project demonstrates practical experience administering a Windows Server 2022 enterprise environment with security, monitoring, and documentation best practices.